

18/08/2025

CUBICLE 5:

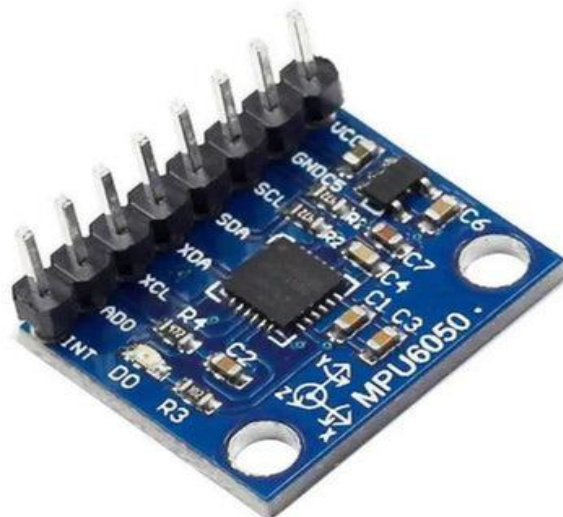
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GY-521



- MPU-6050 chip
 - A 3-axis accelerometer + 3-axis gyroscope
- Built-in Digital Motion Processor (DMP) for motion tracking
- I2C interface (SCL and SDA pins) for communication with Arduino or other microcontrollers

What does it do?

The GY-521 can:

- Measure acceleration (tilt, movement)
- Measure rotation (angular velocity)
- Detect orientation (for balancing robots, drones, etc.)
- Combine data from both sensors using the DMP for better motion tracking

Common Projects Using GY-521:

- Self-balancing robots
- Gesture-controlled devices
- Drones and RC planes
- Fitness trackers or fall detection
- Motion-activated lights or alarms

Arduino Temperature-Controlled Relay with LED (HW-506)



DESCRIPTION:

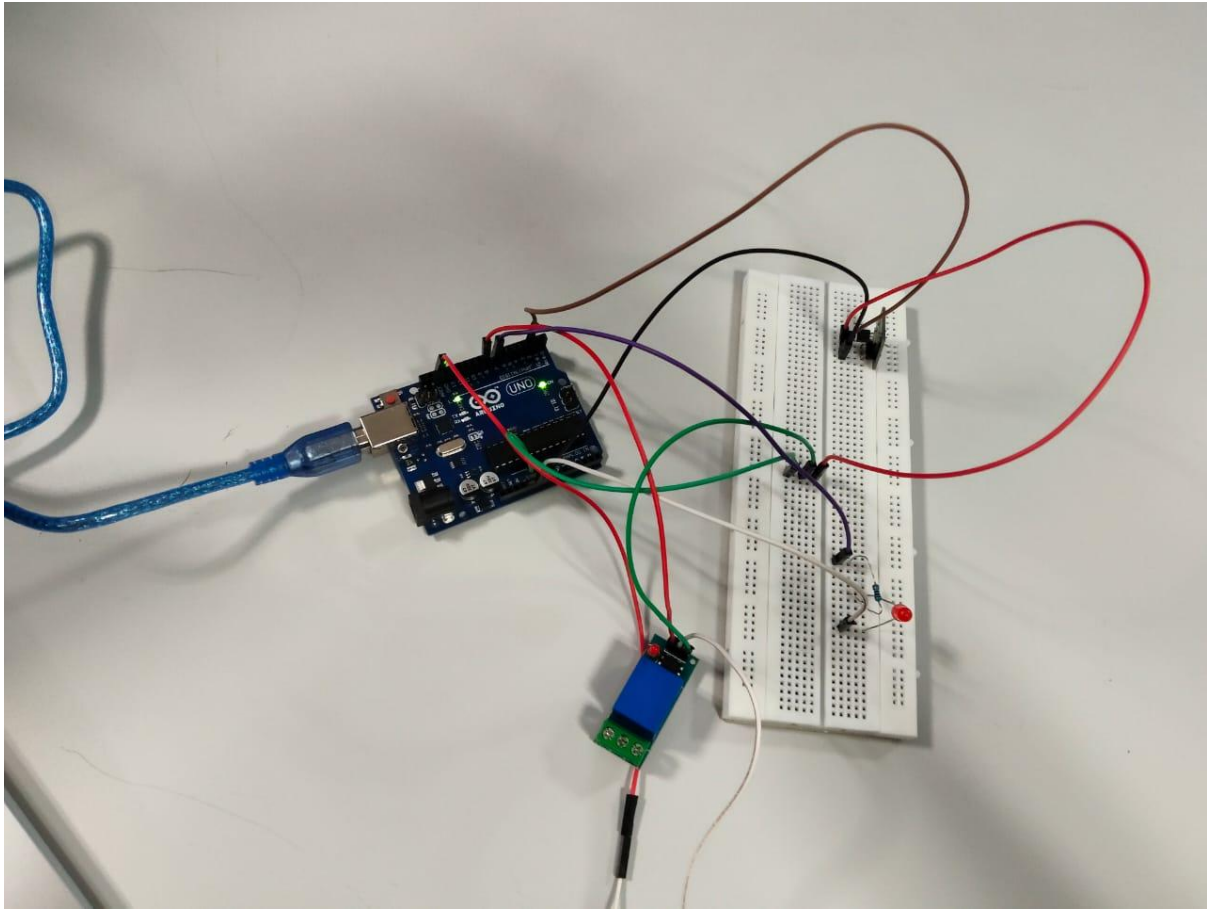
This is a standalone Arduino project that reads temperature using the HW-506 DS18B20 module and controls a relay based on a temperature setpoint.

- When the temperature exceeds the setpoint, the relay turns ON (to drive a fan, heater, or light).
- When the temperature drops below the setpoint, the relay turns OFF.
- An LED indicates relay status: ON when the relay is ON, OFF otherwise.
- No Wi-Fi, Bluetooth, or smartphone required — fully offline.

PIN CONNECTIONS:

Function	Module Pin	Arduino Pin
HW-506 DQ (data)	DQ	D2
HW-506 VCC	VCC	5V
HW-506 GND	GND	GND
Relay IN	IN	D7
Relay VCC	VCC	5V
Relay GND	GND	GND
LED anode (+)	–	D8 (with 220 Ω resistor)
LED cathode (–)	–	GND

BOARD SETUP:



PROGRAM:

```
#include <OneWire.h>

#include <DallasTemperature.h>

// Pins

const int tempPin = 2;  // DS18B20 DQ
const int relayPin = 8;  // Relay control
const int ledPin = 7;  // Status LED
```

```
// Temperature control

const float setPoint = 30.0; // Set temperature in °C


// Setup DS18B20
OneWire oneWire(tempPin);
DallasTemperature sensors(&oneWire);


void setup() {
  Serial.begin(9600);
  sensors.begin();

  pinMode(relayPin, OUTPUT);
  pinMode(ledPin, OUTPUT);

  digitalWrite(relayPin, LOW); // Relay OFF initially
  digitalWrite(ledPin, LOW); // LED OFF initially
}


void loop() {
  sensors.requestTemperatures();
  float tempC = sensors.getTempCByIndex(0);

  Serial.print("Temperature: ");
  Serial.print(tempC);
```

```
Serial.println(" °C");
```

```
if (tempC >= setPoint) {
```

```
    digitalWrite(relayPin, HIGH); // Turn ON relay
```

```
    digitalWrite(ledPin, HIGH); // Turn ON LED
```

```
} else {
```

```
    digitalWrite(relayPin, LOW); // Turn OFF relay
```

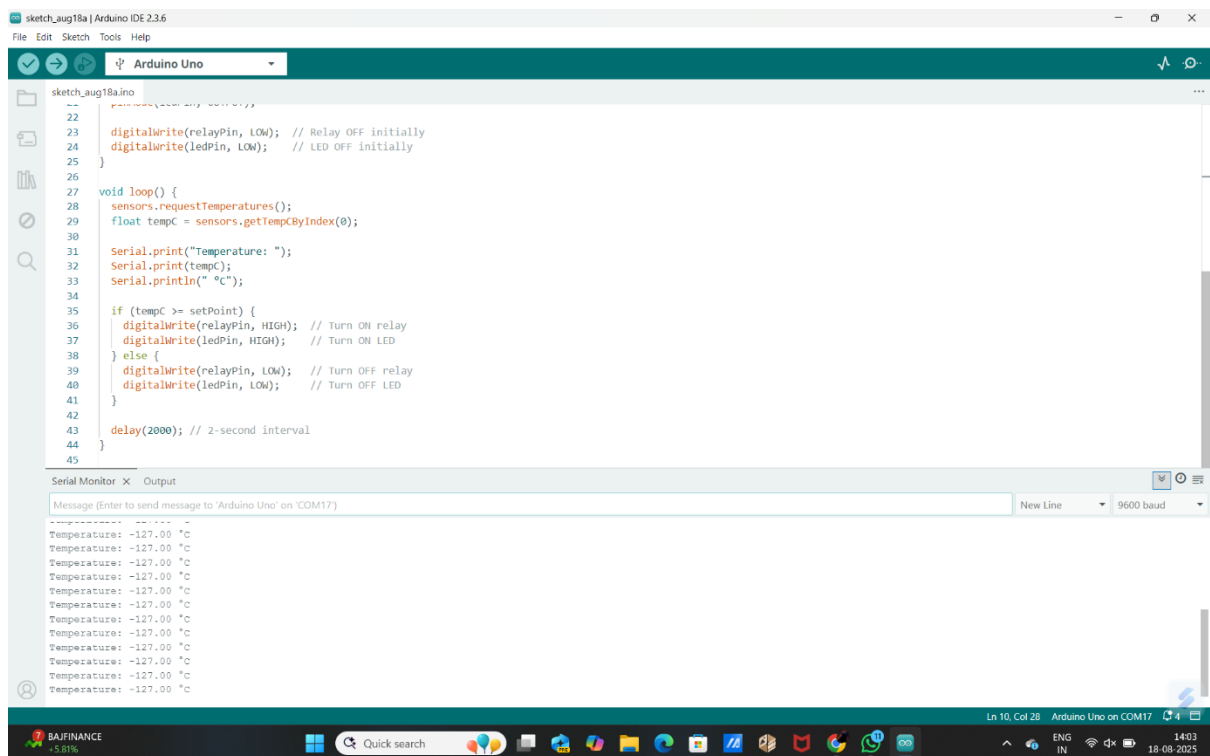
```
    digitalWrite(ledPin, LOW); // Turn OFF LED
```

```
}
```

```
delay(2000); // 2-second interval
```

```
}
```

OUTPUT:



The screenshot shows the Arduino IDE interface. The main editor window displays the code for sketch_aug18a.ino. The code includes comments for initial pin states, a loop function that reads temperature, prints it, and controls a relay and LED based on a set point. The Serial Monitor at the bottom shows the output of the program, displaying the temperature reading: -127.00 °C.

```
sketch_aug18a.ino
22 // Relay OFF initially
23 digitalWrite(relayPin, LOW); // Relay OFF initially
24 digitalWrite(ledPin, LOW); // LED OFF initially
25 }
26
27 void loop() {
28     sensors.requestTemperatures();
29     float tempC = sensors.getTempCByIndex(0);
30
31     Serial.print("Temperature: ");
32     Serial.print(tempC);
33     Serial.println(" °C");
34
35     if (tempC >= setPoint) {
36         digitalWrite(relayPin, HIGH); // Turn ON relay
37         digitalWrite(ledPin, HIGH); // Turn ON LED
38     } else {
39         digitalWrite(relayPin, LOW); // Turn OFF relay
40         digitalWrite(ledPin, LOW); // Turn OFF LED
41     }
42
43     delay(2000); // 2-second interval
44 }
45
```

Serial Monitor X Output

Message (Enter to send message to 'Arduino Uno' on 'COM17')

Temperature: -127.00 °C
Temperature: -127.00 °C
Temperature: -127.00 °C
Temperature: -127.00 °C
Temperature: -127.00 °C
Temperature: -127.00 °C
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Ln 10, Col 28 Arduino Uno on COM17 1403 18-08-2025